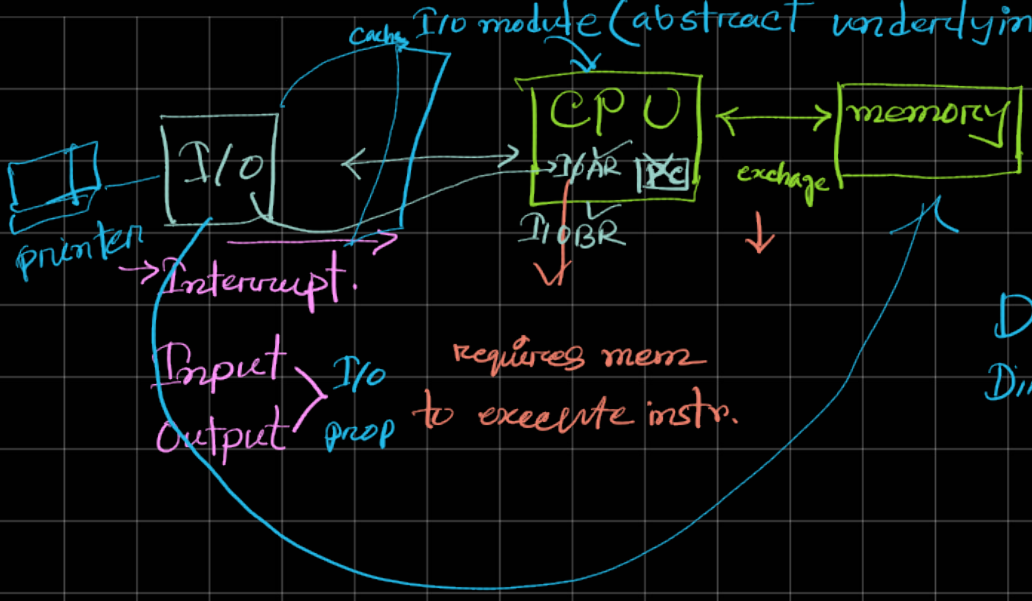


Input / Output → I/O prop requires mem to execute instr.

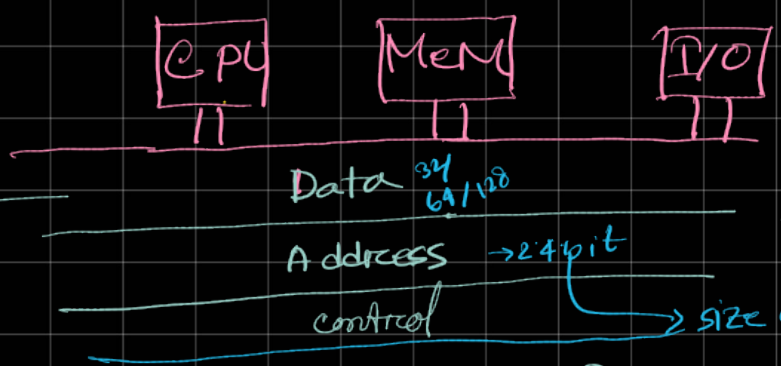
DMA
Direct Mem access

Internal data vs external data.

only from input/output module.

I/O module takes data from input devices.

Types of interconnection



size of RAM : 2^{24}
$= 2^4 + 2^{20}$
= 16 Megabyte.

Bus → single line - single bit. ∴ Timestamp

↓ collection of lines.

1st → CPU obtain Address of I/O device (transfer to control)

size of bus → size of system bus & that size is stored in registers.

64 bit archi → size of Address bus.

control Bus
Mem R/W = I/O

32 bit bus → 64 bit instr
↓
Fetch mem twice.

what if 32 bit instr?
=> Then also 64 bit better as next instr will already be fetched. so m.m access is reduced.

1) can't exchange bit in point to point interconnection

2) has error connection code.

Send to god (I own god)

Green → QPI

Ash → PCIE

Black → BUS